

v2 Phase A - pretraining ablation + real ADNI labels

2026-07-02

v2 Phase A - pretraining ablation (for Yoni)

Follow-up to the v1 results. Two things changed since:

1. **Real ADNI clinical labels.** We replaced the CDR proxy (from Sagi’s manifest) with the **real diagnosis** (ADNI DXSUM: CN / MCI / Dementia) and **real amyloid status** (UC Berkeley amyloid PET), matched by PTID. Coverage over our 215 ADNI subjects: 212 with a diagnosis (CN 77 / MCI 101 / Dementia 34), 188 with amyloid (84 negative / 104 positive). This unblocks Brain-JEPA’s ADNI tasks (NC/MCI, Amyloid) with honest labels.
2. **Phase A pretraining ablation.** Four pretrainings, one factor each, evaluated with the leakage-free linear probe (frozen encoder, 70:30 subject split, metric matched to each SOTA):
 - **base** - reference (learned spatial pos, strided-conv temporal downsample, full patchify)
 - **fourier** - Fourier spatial positional encoding instead of the learned table
 - **noblock2** - drop the final patchify ResBlock (block_2, ~83% of patchify params)
 - **pool** - temporal downsample by average pooling instead of strided conv (+ no block_2)

Results (linear probe on the frozen encoder)

Each cell is the metric that the corresponding SOTA reports for that task. ADNI rows use the **real** labels for base and noblock2 (fourier/pool ADNI were auto-probed earlier with the CDR proxy and are not comparable - marked *).

Axis (metric)	base	fourier	noblock2	pool	SOTA
ABIDE Age (Acc)	0.80	0.71	0.70	0.70	SLIM-Brain 0.64 / SwiFT 0.62
ABIDE Sex (F1)	0.82	0.82	0.78	0.76	LCM 0.87
HCP Sex (Acc)	0.81	0.79	0.84	0.80	SLIM-Brain 0.91 / LCM 0.73
HCP Sex (AUC)	0.90	0.90	0.92	0.89	-
ABIDE Autism (AUROC)	0.53	0.50	0.47	0.57	BNT 0.80 / BrainGFM 0.71
ADNI NC/MCI (AUC)	0.43	0.45*	0.49	0.48*	Brain-JEPA 0.77 (acc)
ADNI AD/HC (AUC)	0.61	0.64*	0.56	0.58*	BrainGFM 0.80
ADNI Amyloid (AUC)	0.53	-	0.51	-	Brain-JEPA 0.71 (acc)

Reading

Two clear messages

- **Demographic / global structure is learned well.** ABIDE Age **0.80**, above the volumetric SOTA (0.62-0.64); HCP Sex 0.84 acc / 0.92 AUC, above LCM (0.73 F1), near SLIM-Brain; ABIDE Sex 0.82 F1. The encoder clearly captures age/sex structure.
- **Clinical signal is at chance under linear probing (with the real labels).** ADNI NC/MCI ~0.43-0.49 AUC, AD/HC ~0.56-0.61, Amyloid ~0.51, Autism ~0.5. The **frozen** encoder does not linearly separate the clinical tasks.

The CDR proxy was misleading (important)

With the CDR proxy, noblock2's ADNI AD/HC looked like 0.76 AUC. With the **real diagnosis** it is **0.56**. So the earlier "clinical win" was an artifact of the proxy correlating with something other than the true diagnosis. **We report only the real-label numbers.**

The architecture ablation did not help

- **pool** is worse or equal everywhere -> dropped.
- **noblock2** trades: better HCP Sex (+0.03 acc, +0.02 AUC) but worse ABIDE (Age -0.10, Sex -0.04); clinical unchanged (chance). It is **~equivalent to base but 6x lighter** (1.8 M vs 10.7 M patchify params) - a size win, not a quality win.
- **base** remains the strongest overall.
- Note: removing 83% of the patchify (block_2) did **not** degrade performance -> the patchify is **not** capacity-limited, so adding patchify depth is unlikely to help. The bottleneck is elsewhere (the mostly-frozen ViT and/or the SSL objective), not the tokenizer.

Conclusion and next step

The linear probe (frozen encoder) is **the weakest downstream protocol**; Brain-JEPA reports its headline numbers by **fine-tuning**. Our frozen-encoder clinical results being at chance is the classic signal that the encoder must **adapt** to the task.

Next experiment (decisive): fine-tuning (point 4). Unfreeze the encoder and train end-to-end on ADNI NC/MCI (Brain-JEPA's headline task), mean +/- std over seeds, same 70:30 test set. If fine-tuning recovers clinical signal -> we move toward Brain-JEPA's 0.77. If it does not -> the SSL pretraining did not learn a recoverable clinical representation, which is itself an honest, reportable result.

Still pending (Brain-JEPA coverage): OASIS-3 AD Conversion (scans in hand; needs the ADRC clinical CDR spreadsheet from oasis-brains.org). Out of reach short-term: UK Biobank (paid DUA), HCP-Aging (NDA), MACC (restricted).